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Data Science

Retrieval-Augmented Vision-Language Agent for CCTVs

Group No. 11

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Project Guide
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Abstract

- CCTV systems generate massive volumes of video daily, making manual review slow, impractical, and prone to missed details.
- Our intelligent system automates surveillance analysis by continuously detecting, isolating, and summarizing significant events.
- Users can interact with the video archive through natural language queries instead of manually scrubbing timelines.
- This transforms raw, unstructured footage into a searchable knowledge base of events.
- Critical incidents can now be found in seconds instead of hours, greatly improving monitoring efficiency.

Sustainable Development Goals(SDG) mapped

SDG 11: Sustainable Cities and Communities
Intelligent surveillance enhances safety and security in commercial spaces.

Introduction

- The immense volume of daily CCTV footage makes manual review for specific incidents impractical.
- Manually searching through hours of footage is a slow, labor-intensive, and error-prone process.
- Basic automated solutions, like motion detection, are unreliable and generate frequent false alarms without providing any event context.
- A critical gap exists in the ability to effectively search video archives for specific, nuanced activities, such as verifying unauthorized entry into a secure room.

Motivation

- Retail industries face major challenges like theft, fraud, and inefficiencies in operations.
- Managers often lack tools to quickly extract key information from large amounts of video data.
- Advances in vision-language AI models provide an opportunity to automate and enhance surveillance analysis.
- To bridge the gap between raw video streams and meaningful insights for decision-making.

Objectives

- Develop a real-time vision pipeline using DeepSORT for accurate person detection and tracking in restricted areas.
- Automate event processing with ESRGAN for image enhancement and a Vision-Language Model to generate JSON summaries.
- Store structured summaries and metadata in ChromaDB to build a searchable event knowledge base.
- Enable natural language querying through a FastAPI backend powered by Mistral 7B for precise event retrieval.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
1	Smart CCTV Video Summarization: A Deep Learning Fusion of YOLO and DeepSORT Algorithms	Vaishali Jabade, Aditya Ingale, Rushikesh Joshi, Gajanan Kulkarni	2024	Proposed a CCTV summarization framework using YOLOv5 for real-time object detection and Deep SORT with Kalman filtering for object tracking. Frames with dynamic elements are merged to produce concise video summaries.	Limited to YOLOv5 and Deep SORT; lacks evaluation on large-scale, diverse real-world datasets.
2	Detection of Criminal Activities Through CCTV by Live Footage Analysis	Rahul Kumar, Shilpi Sharma, Ayan Sar, Tanupriya Choudhury	2024	Developed a YOLO-based CCTV crime detection system for theft,vandalism, and violent behavior. Integrated deep learning with real-time alerting, transfer learning, and data augmentation for adaptability	High computational demand; privacy and ethical concerns; performance may drop in poor lighting or crowded scenes.
3	Video Summarization Techniques: A Comprehensive Review	Toqa Alaa, Ahmad Mongy, Assem Bakr, Mariam Diab, Walid Gomaa	2024	Survey of extractive and abstractive summarization methods, including clustering, deep learning (CNN, RNN, LSTM, Transformers), attention, GANs, reinforcement learning, and multi-modal approaches.	Being a review, no implementation; highlights research gaps such as dataset limitations and real-world deployment issues.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
4	Recent Challenges and Opportunities in Video Summarization with Machine Learning Algorithms	Payal Kadam, Shruti Patil, Deepali Vora, Sashikala Mishra, Ketan Kotecha, Ajith Abraham, Lubna Abdelkareim Gabralla	2022	Comprehensive survey of video summarization methods from 2005–2022. Categorized techniques into static vs dynamic, supervised vs unsupervised, single-view vs multi-view. Discussed datasets (TVSum, SumMe) and metrics (F-score, precision, recall). Highlighted need for robust MVS approaches and advanced deep architectures like3D-CNNs and Conv-LSTMs.	Survey only, no implementation; MVS methods remain underexplored and face synchronization and real-world challenges.
5	Automatic Video Summarization with Timestamps using Natural Language Processing Text Fusion	Ahmed Emad, Fady Bassel, Mark Refaat, Mohamed Abdelhamed, Nada Shorim, Ashraf AbdelRaouf	2021	Proposed a multimodal system combining three modalities: visual analysis (frames), facial emotion recognition (CNN), and audio-to-text transcription. Outputs fused via NLP to generate a coherent textual summary with timestamps. Human evaluation achieved ~87% agreement on summary accuracy.	Dependent on quality of NLP fusion; struggles with silent or emotionless scenes; requires further refinement for natural sentence flow.
6	Intelligent Video Surveillance Based on YOLO: A Comparative Study	Rakesh Garg, Someet Singh	2021	Compared YOLOv3, YOLOv4, and YOLOv5 for handheld weapon detection in surveillance. YOLOv4 achieved the highest mAP (98.79%) and recall. Also tested tiny versions for real-time deployment.	YOLOv3/v4 are computationally heavy; tiny models reduce accuracy; lacks testing in varied environmental conditions.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
7	Unsupervised Video Summarization with a Convolutional Attentive Adversarial Network	Guoqiang Liang, Yanbing Lv, Shucheng Li, Shizhou Zhang, Yanning Zhang	2021	Uses a generator with fully convolutional sequence and attention mechanism; discriminator distinguishes summary-weighted features via GAN loss	Additional complexity from attention layers; potential overfitting on small datasets.
8	YOLO-Based Video Processing for CCTV Surveillance	Sujata Terdal, Sayyada F., Amulya Reddy, Manasi Koppal, Ameena Fatima	2020	Developed real-time CCTV surveillance using YOLOv3 for hazardous/suspicious object detection. Preprocessing included denoising, grayscale conversion, and resizing. Achieved 98.3% detection accuracy with high precision and recall (1.00 for "Person").	Focuses only on object detection, not activity recognition; scalability in crowded or multi-camera setups not addressed.
9	YOLO Based Real-Time Human Detection for Smart Video Surveillance at the Edge	Huy Hoang Nguyen, Van Truong Bui, Thi Nhung Ta, Hung Manh Pham	2020	Proposed a lightweight YOLOv2-based human detector optimized with residual blocks and SPP for edge devices. Achieved ~95% accuracy on INRIA and PENN-Fudan datasets; 2 FPS on Raspberry Pi 3B.	Limited processing speed (2 FPS); trade-off between accuracy and speed may restrict real-time use in high-density scenarios.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
10	Comprehensive Video Understanding: Video Summarization with Content-Based Video Recommender Design	Yudong Jiang, Kaixu Cui, Bo Peng, Changliang Xu	2019	Proposed a deep neural network (HighlightNet) treating video summarization as a content-based recommender problem. Integrated raw, semantic, audio, and vision features with GRU modeling, self-supervised learning, and data augmentation.	Requires large labeled datasets; subjectivity of summaries not fully addressed; risk of overfitting with limited data.
11	Cycle-SUM: Cycle-Consistent Adversarial LSTM Networks for Unsupervised Video Summarization	Li Yuan, Francis E. H. Tay, Ping Li, Li Zhou, Jiashi Feng	2019	Bi-LSTM selector + cycle-consistent dual GAN evaluator to preserve mutual information between summary and original video	More complex architecture; may require extensive training and tuning.
12	Deep Learning Approach for Suspicious Activity Detection from Surveillance Video	C. Rami Reddy, A. Sreenivasulu	2018	Implemented CNNs for feature extraction and LSTMs for temporal analysis to detect suspicious activities in CCTV. Used VGG-16 pretrained on ImageNet for behavior inference.	Tested on limited campus datasets; real-world scalability and accuracy in complex environments not validated.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
13	Unsupervised Video Summarization with Adversarial LSTM Networks	Behrooz Mahasseni, Michael Lam, Sinisa Todorovic	2017	GAN framework with LSTM autoencoder summarizer and LSTM discriminator for unsupervised keyframe selection	May generate visually faithful but semantically weak summaries, struggles with long/complex videos and subjective variations.
14	Rapid Video Summarization on Compressed Video	Jurandy Almeida, Ricardo da S. Torres, Neucimar J. Leite	2010	Utilizes compressed-domain indicators such as I-frame DC images and HSV histograms to generate summaries without fully decoding video.	Performance degrades in scenes with dynamic or complex backgrounds; limited by reliance on compressed-domain features.
15	A Video Summarization Approach Based on Machine Learning	Wei Ren, Yuesheng Zhu	2008	Developed a machine learning model for automatic video transition prediction using features like cut, fade, dissolve, and pixel likelihood ratio. Keyframes are selected via a coarse-to-fine strategy to improve summarization.	Limited feature set; performance depends on shot transitions, less effective for long panning or tilting shots.

Research Gap (Limitations of existing systems)

- Limited testing on small or synthetic datasets, lacking real-world retail validation.
- High computational demand restricts real-time and edge deployment.
- Accuracy drops in low-light, crowded, or multi-camera environments.
- Focus mainly on object detection, not on activities or context.
- Generated summaries are often incomplete or incoherent.
- No support for intelligent natural language querying of incidents.
- Scalability issues for large-scale, real-time retail use.

Problem Definition

- The massive volume of CCTV footage makes manual review inefficient and unsustainable.
- Existing systems lack intelligent search, requiring operators to manually locate specific activities like unauthorized entry.
- Basic motion detection provides limited context, causing false positives and alert fatigue.
- Delays between incidents and discovery hinder proactive response and timely evidence collection.

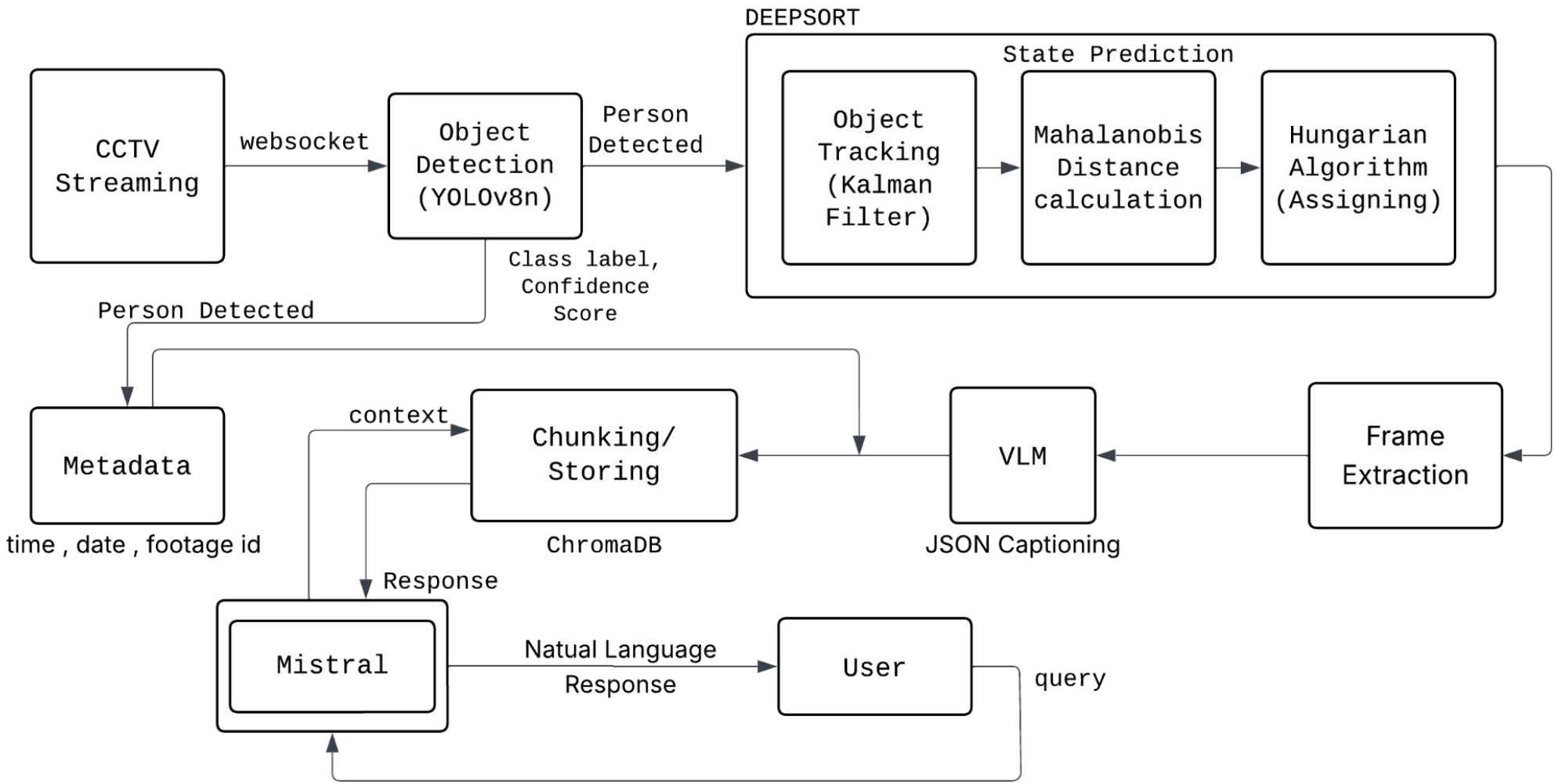
Scope

- Real-time logging of activities happening in safe rooms (Exam Strong room, High security rooms etc)
- Human detection and summarization to automatically generate concise, context-aware descriptions.
- Semantic storage of json data in a vector database with metadata for long-term retrieval.
- Natural language querying of surveillance data through an AI-powered LLM assistant.
- Support for incident investigation by providing evidence-backed answers instantly.

Technological Stack

1. Real-Time Detection & Tracking
   - DeepSORT: Primary framework for robust, real-time multi-object tracking.
   - Kalman Filter & Hungarian Algorithm: Ensures accurate, consistent tracking by predicting object movement and matching detections across frames.
1. Image Processing & VLM Summarization
   - OpenCV: Handles the ingestion and processing of video frames from CCTV feeds.
   - ESRGAN: Upscales frame resolution to enhance image quality and preserve critical details
   - VLM (Smol VLM/Moondream2/Phi-3 Vision): Analyzes enhanced frames and generates structured JSON summaries of detected events.
1. Knowledge Base & Natural Language Querying
   - ChromaDB: Stores JSON summaries and embeddings, enabling efficient search and retrieval.
   - LLM (Mistral 7B): Interprets user queries and synthesizes context-aware answers from retrieved summaries.
   - FastAPI: Backend API managing user queries, coordinating retrieval and LLM responses.

Proposed system architecture/Working/Module Diagram



Prototype Design Demonstration

Our prototype integrates a series of specialized AI models and libraries to create a complete, end-to-end pipeline that transforms raw video into a searchable intelligence source. The design is broken down into three core modules:

- **Capture & Detection Module**: This is the "eyes" of our system. It uses OpenCV to ingest a video feed in real-time. On each frame, the DeepSORT algorithm, with its integrated Kalman Filter and Hungarian algorithm, is used to detect and assign a persistent tracking ID to any person in the frame. The output of this module is a stream of video frames where any detected individuals are flagged and tracked.

- **Analysis & Summarization Module**: Once a person is detected, this module takes over. It uses OpenCV functions and an ESRGAN model to enhance the quality of the captured frames, ensuring maximum details. These enhanced frames are then fed into a Vision-Language Model (VLM), such as SmolVLM, Moondream2 or Phi-3 Vision. The VLM's sole purpose is to analyze the visual information and generate a structured JSON summary that describes the activity taking place.

- **Query & Interaction Module**: This module makes the data useful. The JSON summaries are stored in a ChromaDB vector database. A FastAPI backend exposes an API endpoint for users to submit queries. When a query is received, it's used to retrieve the most relevant summaries from the database. This context, along with the original query, is passed to a Mistral 7B Large Language Model (LLM), which generates a final, human-readable answer.

Prototype Design Demonstration

End-to-end workflow on a controlled test case.

- **Setup**: The demonstration begins by running our Python application and pointing it to a pre-recorded video file depicting a secure room—our target environment. This controlled setup allows us to reliably test and showcase the system's functionality.

- **Real-Time Trigger and Analysis**: As the video plays, the system is initially idle. When a person enters the frame in the video, the console immediately shows the **DeepSORT** algorithm detecting and assigning a tracking ID to the person. Seconds later, the system displays the **JSON summary** generated by the VLM for that specific event, proving that the automated detection and summarization pipeline is fully functional.

- **Natural Language Query**: With the event now detected and stored, we demonstrate the core innovation. Using a simple interface or command-line tool, we send a natural language query to our **FastAPI** endpoint, asking, "Did anyone enter the room just now and what were they doing?".

- **Verification**: The system's final output is displayed: a natural language sentence generated by the **Mistral 7B LLM**, accurately describing the event based on the VLM summary it just retrieved. This successful query-to-answer loop serves as the proof of concept, showing the working of the prototype.

Implementation Status

- Object Detection and Live tracking.
- Image enhancement pipeline using OpenCV and ESRGAN Neural Network Inference.
- Websocket based image transmission.
- Collection of Raw images consisting of a closed space to test scenarios.
-

Research paper Status

I2ITCON (hope foundation) - Applied (Not Accepted)
Paper ID : 4741
Title: A Retrieval-Augmented Framework for Semantic Video Surveillance using Vision-Language Models

link : https://i2itcon.isquareit.edu.in/

ICICIS 2026 (Ajeenkya DY Patil UNI) - Applied
Paper ID : 204
Title: A Retrieval-Augmented Framework for Semantic Video Surveillance using Vision-Language Models

Patent/Copyright filing Status

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Thank You...!!